

D1 Technical Report - Streaming Learner & AutoML Note

CSAI415 Special Topics in Artificial Intelligence | Climate Evidence GraphRAG Agent

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1. D1 scope and data

We completed D1 Track A and the required River component. The retrieval experiment uses 300 open-access climate PDFs, 49,541 page-aware chunks, and a repaired 120-question page-grounded proxy benchmark. The first auto-generated benchmark was preserved as a legacy draft because several labels were not anchored to the page named in the question; the repaired set uses exact source-page evidence and is still flagged for manual review.

2. AutoML Track A - supervised hybrid kNN retriever

The runnable AutoML pipeline combines BM25 lexical retrieval with TF-IDF features reduced by TruncatedSVD and kNN search. Optuna ran 30 trials on an 84-question training split and was evaluated on a held-out 36-question test split. Search space: $k=\{5,10,15,20\}$, $\text{metric}=\{\text{cosine}, \text{euclidean}, \text{manhattan}\}$, $\text{SVD dim}=\{50,100,200,300\}$, $\text{normalization}=\{\text{none}, \text{minmax}, \text{zscore}\}$, and hybrid lexical weight in $[0,1]$.

Objective: We used a combined AutoML objective that prioritizes ranking quality through NDCG@5 , also rewards Recall@5 , and adds a small latency penalty so the selected retriever is accurate but still practical for interactive use.

$\text{NDCG@5} + 0.25 * \text{Recall@5} - 0.0005 * \text{p95 latency}$.

System	Recall@5	NDCG@5	MRR	p95 latency
Baseline	0.667	0.522	0.493	674.5 ms
AutoML tuned	0.833	0.682	0.645	676.2 ms

Winning configuration: $k=20$, cosine metric, SVD dim=300, min-max normalization, hybrid lexical weight=0.815. **Impact:** held-out Recall@5 increased by 0.166 and NDCG@5 by 0.160 with almost unchanged latency.

3. River online learner with ADWIN

We implemented query-to-topic classification with River Bag-of-Words + Multinomial Naive Bayes over six climate topics. A 600-query stream injected concept drift at query 350. Evaluation followed the correct sequential order: predict -> score -> update ADWIN on the error stream -> learn_one(). ADWIN first alerted at query 377.

Metric	Result
Pre-drift accuracy	89.97%
Drift-window accuracy	2.22%
Post-drift accuracy	95.65%
Final cumulative / rolling	78.33% / 100%

Figure 1. Rolling accuracy drops after drift at query 350; ADWIN alerts at 377 and 473; performance then recovers.

4. Decisions and pitfalls

- Kept Track A because it directly matches the brief and gives a reproducible supervised tuning path.
- Used page-level evaluation because D1 questions are page-grounded and citations must later point to page ranges, not arbitrary chunk fragments.

- High lexical weight indicates exact titles, policy terms, and climate vocabulary remain important; dense search alone was weaker.
- Main pitfall: the first auto-generated gold set produced misleading scores, so we replaced it with a repaired page-grounded proxy set and disclose that it still needs manual review.
- Latency rose only 1.7 ms at p95 for the tuned retriever, but later deliverables should reduce CPU latency and replace proxy labels with a manually checked benchmark.

5. Reproducible D1 artefacts

Category	Files
Runnable scripts	src/retrieval/automl_tuner.py src/learning/river_topic_classifier.py
Winning run card	configs/run_card_d1.yaml
Result files	reports/tables/d1_baseline_vs_automl_metrics.csv reports/tables/d1_automl_trials.csv reports/tables/prequential_online_learning_results.csv reports/figures/prequential_accuracy_plot.png